London Travel Demand Survey sampling and design

Introduction

The London Travel Demand Survey (LTDS) is a sample survey of London households carried out annually since 2005/06. The survey is conducted via face-to-face interviews in people's homes at a randomly selected sample of addresses in London. The objective of the survey is to quantify the travel patterns of London residents and how these vary for different population groups. The overall target is to complete 8,000 household interviews in each year of the survey (running from 1 April to 31 March).

This document sets out a brief overview of the survey and sample design. For more information please contact LTDSenquiries@tfl.gov.uk.

Survey design

Background

The content and design of the LTDS survey is similar to the household survey that was carried out as part of the London Area Transport Survey (LATS) in 2001, which developed from surveys carried out in 1971, 1981 and 1991. Each survey collected details of travel by individual members of each sample household on a random day. LATS covered weekdays only, while for LTDS travel days are chosen to give an even distribution across all seven days of the week.

The change from a decennial series of surveys to a rolling programme was made to meet the changing needs for the accurate monitoring of trends in travel demand and travel behaviour more regularly.

Uses of the LTDS data

The LTDS is designed to derive year-on-year estimates of travel and transport indicators, such as London residents' mode shares and trip rates, with some breakdown by area, sub-region and population sub-groups. The survey is not intended to monitor trends for shorter periods than a year. For estimates at borough level and below it is usually necessary to combine the sample achieved over more than one year to ensure the necessary sample robustness and data accuracy. The LTDS results are not representative of all travel in London since travel by commuters and visitors is not captured.

The LTDS data is widely used within TfL and the London boroughs and plays an essential role in the development and monitoring of policy initiatives and the Mayor's Transport Strategy. Specifically, the LTDS contributes contextual and tracking data for a number of high-profile transport policy areas including travel demand assessment and forecasting, cycling and walking policy and the measurement of mode shares. The spatial data acquired is used to underpin sub-regional and borough-level analysis, to target efforts and to assess performance against the Local Implementation Plan (LIP) targets.

Key trends from the LTDS is published in the annual Travel in London reports.

The interview

An initial **household interview** takes place with a responsible adult to collect basic information about the household and to establish the number of household members and their demographic information. Data on household vehicles is also collected at the household interview.

This is followed by an **individual interview** with each household member aged five and over, which includes a one-day interviewer-administered travel diary. The **travel diary** interview collects detailed information about all the journeys each respondent made on a pre-assigned travel day, usually one or two days before the interview date.

Sample selection

Background

The LTDS is based on a random sample of households drawn from Royal Mail's Postcode Address File for each survey year. The sample is designed so that it is representative both across London and for each sub-region.

Sample structure

The survey area is divided into 528 Sample Control Areas (SCAs). The main function of an SCA is to provide suitable areas in which interviewers can carry out their tasks. SCAs were first defined for the 2001 LATS, based on combinations of 1991 Census Enumeration Districts. For the LTDS these were redefined using the 2001 Census Output Areas to provide the closest possible match to the LATS SCA definitions while maintaining the requirement to nest within borough boundaries and to make them uniform in size of population. This resulted in the creation of 528 LTDS SCAs which have remained the same since the first year of the LTDS (2005/06).

Calculation of stint sizes

The number of addresses issued in each SCA (stint size) is calculated using a methodology which is designed to achieve roughly 15.2 interviews in each SCA, thus spreading the desired 8,000 interviews uniformly between the 528 SCAs. The calculation of stint sizes also reflects a strike rate which is calculated based on the number of interviews achieved against the target in each SCA in previous years.

Expansion of the data

Weighting

The objective of the weighting process is to derive expansion factors that weight the data to the Greater London population.

The household data is weighted to reflect the number of households by household structure type at the borough level. The person sample is aggregated by gender and age and weighted to match population estimates of gender and age by borough.

Four additional expansion factors are calculated, including weekday and term-time weights at the household and person level.

Use of estimates and confidence intervals

Given that the LTDS is based on a sample population, the estimates produced differ from the true value in the population. The difference between these values is due to statistical sampling error, which is a measure of the extent to which an estimate may depend on the particular sample chosen. In general, large samples give more precise estimates with lower sampling error compared to small samples. Sample sizes limit the extent to which analysis can be carried out for small population groups or local areas.

As a general rule, estimates based on fewer than 500 households or individuals should be treated cautiously, and those based on fewer than 100 households or individuals should not be used. For most practical purposes, the LTDS analyses are limited to borough level or greater, except where multiple years of data have been combined to boost the sample size.

Table 1 below shows the approximate 95 per cent confidence intervals for a mode share estimate (p) given the stated sample of trips (n).

Table 1 Indicative confidence intervals for mode share estimates based on differing samples of trips.

Sample size of trips (n)	p=5 per cent	p=25 per cent	p = 50 per cent
500	5	11	12
1,000	4	8	9
2,000	3	5	6
3,000	2	4	5
4,000	2	4	4

For example, a mode share of about 25 per cent for a borough with a sample of 2,000 trips will be estimated with a confidence interval of ±5 per cent.